OIL PUMP

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Clearance Over Rotors (Max.)	0.10 mm	0.004 in.
Cover Out-of-Flat (Max.)	0.025 mm	0.001 in.
Inner Rotor Thickness (Min.)	10.699 mm	0.421 in.
Outer Rotor Thickness (Min.)	10.699 mm	0.421 in.
Outer Rotor Clearance (Max.)	0.039 mm	0.015 in.
Outer Rotor Diameter (Min.)	85.924 mm	3.383 in.
Tip Clearance Between Rotors (Max.)	0.20 mm	0.008 in.

OIL PRESSURE

DESCRIPTION	SPECIFICATION	
	Metric	Standard
At Curb Idle Speed*	25 kPa	4 psi
At 3000 rpm	170 - 550 kPa	25 - 80 psi
CAUTION: *If pressure is ZERO at curb idle, DO NOT run engine at 3000 rpm.		

SPECIFICATIONS - TORQUE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Bolts, Balance Shaft Carrier to Block	54	40	-
Fastener - Double Ended, Balance Shaft Gear Cover	12	-	105
Bolt, Balance Shaft Sprocket	28	-	250
Bolts, Balance Shaft Chain Tensioner	12	-	105
Bolts, Balance Shaft Carrier Cover	12	-	105
Bolt, Camshaft Sprocket	115	85	-
Bolts, Connecting Rod Cap	27 +¼ turn	20 +¼ turn	-
Bolts, Crankshaft Main Bearing Cap/Bedplate			
M8 Bolts	Refer to procedure		
M11 Bolts	Refer to procedure		
Crankshaft Damper	136	100	-
Bolts, Cylinder Head	(Refer to 9 - ENGINE/ CYLINDER HEAD - INSTALLATION)		
Bolts, Cylinder Head Cover	12	-	105
Flex Plate to Crankshaft	95	70	-
Flywheel Mounting Bolts	81	60	-
Bolts, Engine Mount Bracket Right	61	45	-
Bolts, Engine Mounting	(Refer to 9 ENGINE/ ENGINE MOUNTING)		
Bolts, Exhaust Manifold to Cylinder Head	23	-	200
Bolts, Exhaust Manifold Heat Shield	12	-	105
Bolts, Intake Manifold - Lower	28	-	250
Oil Filter	20	15	-
Bolts, Oil Pan	12	-	105
Plug, Oil Pan Drain	27	20	-
Bolts, Oil Pump to Block	28	-	250
Bolts, Oil Pump Cover Plate	12	-	105
Bolt, Oil Pump Pick-up Tube	28	20	-
Cap - Oil Pump Relief Valve	41	30	-
Spark Plugs	18	13	-
Bolts, Timing Belt Covers			
- Front Covers to Rear Covers	12	-	105
- Rear Cover	12	-	105
Bolts, Timing Belt Tensioner Assembly	61	45	-